

CHIRAG VIJAY PATIL

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EDUCATION

Virginia Tech, Blacksburg

Aug 2023 - May 2025

Master of Science, Computer Engineering

GPA: 3.96/4

Relevant Coursework: Data Analytics, Computer Vision, Large Scale Software Engineering, Advanced Machine Learning and Deep Learning, Distributed Systems, Database Management Systems, Object Oriented Programming

WORK EXPERIENCE

Software Development Engineer

East Palo Alto, CA, United States

Amazon Web Services, Amazon

Aug 2025

- Contributed to a reliability-critical **AWS** infrastructure team responsible for maintaining high availability across **thousands of RDS hosts** supporting production database workloads.
- Led the migration of system tests from a legacy framework to a modern testing framework, reducing test flakiness by **30–50%** and helping drive the pipeline toward **near zero test failures** and this **saved 100 hours of developer time per month** previously spent on investigating flaky tests.
- Owned the migration of host monitoring and telemetry service from **JDK 8** to **JDK 17**, modernizing services running on long-lived production hosts and enabling **30% performance improvements** through runtime optimizations.
- Collaborated with cross-functional engineers to deploy and validate changes in a production environment, ensuring **zero customer visible downtime** while maintaining strict availability and reliability SLAs.

Graduate Research Assistant

Blacksburg, VA, United States

Li Lab of Applied Machine Learning in Genomics and Phenomics

Sep 2024 - Aug 2025

- Led the development of a fine-tuned **multimodal** Grounded DINO (GDINO) model, leveraging **vision transformers** for precise object detection in confocal root images to enhance the accurate identification of plant root cells, achieving a **20% improvement** in detection accuracy.
- Integrated the Segment Anything Model (microSAM) to refine segmentation accuracy, reducing segmentation errors by **30%** and enhancing object localization.
- Worked on a **hyperspectral dataset** with over **500,000 data points** to enable the effective use of the Segment Anything Model (SAM) for improved segmentation, resulting in a **25% increase** in segmentation precision.

Cyber Security Intern

Mumbai, India

Tele-Networks Technologies Pvt. Ltd

Jan 2022 - Feb 2022

- Improved network performance and security by **30%** through in-depth information gathering using DNS, SNMP, and SMTP, collaborating with cross-functional IT teams to enhance security measures.

PROJECTS

Real-Time E-commerce Analytics & Dynamic Pricing | Kafka, React

Jan 2025 – May 2025

- Designed and implemented a real-time e-commerce pipeline leveraging **React**, **FastAPI**, **Apache Kafka** (KRaft mode), and Apache Spark Streaming to capture user purchase events, apply dynamic surge pricing based on live demand, and synchronize updates across the system with **1.5s end-to-end latency**.
- Engineered a scalable, full-stack architecture using **SQLite**, FastAPI to ensure low-latency data flow and real-time UI updates, achieving **100+ events/sec** throughput and demonstrating effective synchronization between all layers.

Stock Portfolio Management System | Django, JWT, AWS, Docker, Git, Apache, Nginx, Redis, Celery

Aug 2024 – Dec 2024

- Implemented OAuth2 using JWT for Google authentication and secured communication with HTTPS, improving security and preventing cross-site forgery.
- Designed a distributed task scheduling system using **Celery** and **Redis**, reducing server load by **40%** and enabling real-time price tracking with user notifications via SMTP, thus improving user engagement

Brain Tumor Semantic Segmentation using Deep Learning | Python, PyTorch, OpenCV, scikit-learn

Jan 2024 – May 2024

- Implemented **deep learning models**, including **U-Net and modified U-Net** architectures with attention mechanisms, for semantic segmentation of brain tumors from multimodal MRI scans.
- Contributed to a comparative analysis of different methodologies, including traditional object detection (YOLOv9) and transfer learning approaches, for brain tumor detection and segmentation.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, HTML/CSS, C++, Matlab, Go, Ruby, Rust

Cloud & Backend: AWS, PostgreSQL, Django, React.js, Node.js, Flask, Apache Spark, Kafka

Developer Tools & ML: PyTorch, TensorFlow, Docker, scikit-learn, Linux, OpenCV, Celery, Redis, Jira, Git